

LAB PROGRAMME

CHAPTER 1 : INTRODUCTION

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1. Given an array of strings words, return the first palindromic string in the array. If there is no such string, return an empty string "". A string is palindromic if it reads the same forward and backward.

Example 1:

Input: words = ["abc","car","ada","racecar","cool"]

Output: "ada"

Explanation: The first string that is palindromic is "ada".

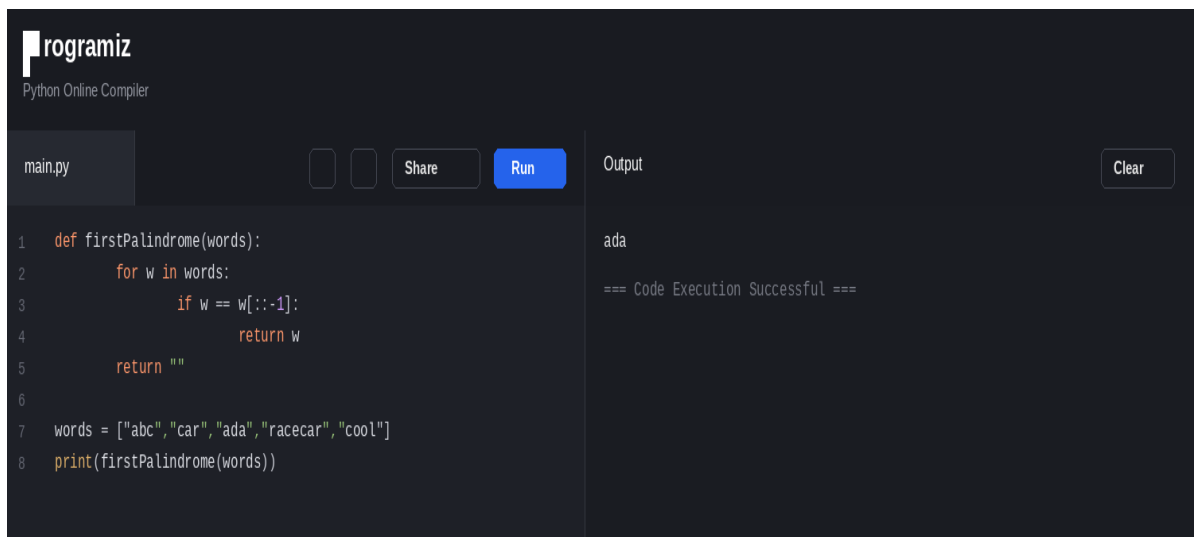
Note that "racecar" is also palindromic, but it is not the first.

Example 2:

Input: words = ["notapalindrome","racecar"]

Output: "racecar"

Explanation: The first and only string that is palindromic is "racecar".



The screenshot shows a web-based Python IDE. On the left, there's a file explorer with 'main.py' selected. Below it, the code editor contains the following Python code:

```
1 def firstPalindrome(words):
2     for w in words:
3         if w == w[::-1]:
4             return w
5     return ""
6
7 words = ["abc","car","ada","racecar","cool"]
8 print(firstPalindrome(words))
```

On the right side, there's an 'Output' section. It displays the result 'ada' and a message '=== Code Execution Successful ==='. There are also buttons for 'Share', 'Run', and 'Clear'.

2. You are given two integer arrays nums1 and nums2 of sizes n and m, respectively. Calculate the following values: answer1 : the number of indices i such that nums1[i] exists in nums2. answer2 : the number of indices i such that nums2[i] exists in nums1 Return [answer1,answer2].

Example 1:

Input: nums1 = [2,3,2], nums2 = [1,2]

Output: [2,1]

Explanation:

Example 2:

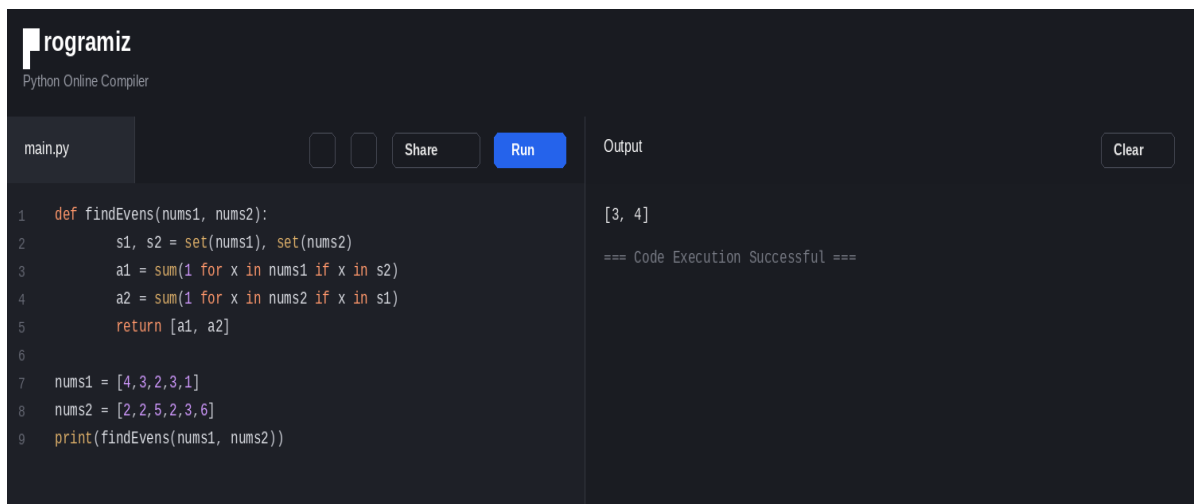
Input: nums1 = [4,3,2,3,1], nums2 = [2,2,5,2,3,6]

Output: [3,4]

Explanation:

The elements at indices 1, 2, and 3 in nums1 exist in nums2 as well. So answer1 is 3.

The elements at indices 0, 1, 3, and 4 in nums2 exist in nums1. So answer2 is 4.



```
1 def findEvens(nums1, nums2):
2     s1, s2 = set(nums1), set(nums2)
3     a1 = sum(1 for x in nums1 if x in s2)
4     a2 = sum(1 for x in nums2 if x in s1)
5     return [a1, a2]
6
7 nums1 = [4, 3, 2, 3, 1]
8 nums2 = [2, 2, 5, 2, 3, 6]
9 print(findEvens(nums1, nums2))
```

Output

[3, 4]

=== Code Execution Successful ===

3. You are given a 0-indexed integer array `nums`. The distinct count of a subarray of `nums` is defined as: Let `nums[i..j]` be a subarray of `nums` consisting of all the indices from `i` to `j` such that $0 \leq i \leq j < \text{nums.length}$. Then the number of distinct values in `nums[i..j]` is called the distinct count of `nums[i..j]`. Return the sum of the squares of distinct counts of all subarrays of `nums`. A subarray is a contiguous non-empty sequence of elements within an array.

Example 1:

Input: `nums = [1,2,1]`

Output: 15

Explanation: Six possible subarrays are:

[1]: 1 distinct value

[2]: 1 distinct value

[1]: 1 distinct value

[1,2]: 2 distinct values

[2,1]: 2 distinct values

[1,2,1]: 2 distinct values

The sum of the squares of the distinct counts in all subarrays is equal to $12 + 12 + 12 + 22 + 22 + 22 = 15$.

Example 2:

Input: `nums = [1,1]`

Output: 3

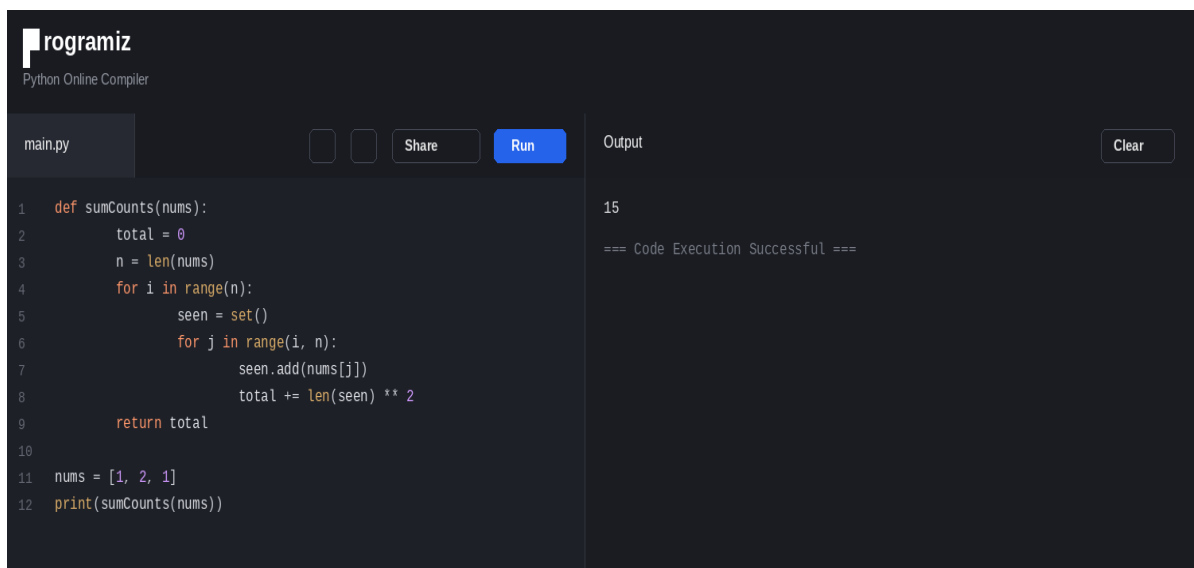
Explanation: Three possible subarrays are:

[1]: 1 distinct value

[1]: 1 distinct value

[1,1]: 1 distinct value

The sum of the squares of the distinct counts in all subarrays is equal to $12 + 12 + 12 = 3$.



The screenshot shows a Python Online Compiler interface. On the left, the code in `main.py` is as follows:

```
1 def sumCounts(nums):
2     total = 0
3     n = len(nums)
4     for i in range(n):
5         seen = set()
6         for j in range(i, n):
7             seen.add(nums[j])
8             total += len(seen) ** 2
9     return total
10
11 nums = [1, 2, 1]
12 print(sumCounts(nums))
```

On the right, the Output panel shows the result:

```
15
=== Code Execution Successful ===
```

4. Given a 0-indexed integer array `nums` of length `n` and an integer `k`, return *the number of pairs (i, j) where $0 \leq i < j < n$, such that `nums[i] == nums[j]` and $(i * j)$ is divisible by `k`.*

Example 1:

Input: `nums = [3,1,2,2,2,1,3]`, `k = 2`

Output: 4

Explanation:

There are 4 pairs that meet all the requirements:

- `nums[0] == nums[6]`, and $0 * 6 == 0$, which is divisible by 2.
- `nums[2] == nums[3]`, and $2 * 3 == 6$, which is divisible by 2.
- `nums[2] == nums[4]`, and $2 * 4 == 8$, which is divisible by 2.
- `nums[3] == nums[4]`, and $3 * 4 == 12$, which is divisible by 2.

Example 2:

Input: `nums = [1,2,3,4]`, `k = 1`

Output: 0

Explanation: Since no value in `nums` is repeated, there are no pairs (i, j) that meet all the requirements.

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main.py ☐ ☐ Share Run Output Clear

```
1 def countPairs(nums, k):
2     count = 0
3     n = len(nums)
4     for i in range(n):
5         for j in range(i + 1, n):
6             if nums[i] == nums[j] and (i * j) % k == 0:
7                 count += 1
8     return count
9
10 nums = [3, 1, 2, 2, 2, 1, 3]
11 k = 2
12 print(countPairs(nums, k))
```

4

=== Code Execution Successful ===

5. Write a program FOR THE BELOW TEST CASES with least time complexity

Test Cases: -

- 1) Input: {1, 2, 3, 4, 5} Expected Output: 5
- 2) Input: {7, 7, 7, 7, 7} Expected Output: 7
- 3) Input: {-10, 2, 3, -4, 5} Expected Output: 5

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```
1 def findMax(arr):
2     # O(n) time, O(1) space - single pass
3     result = arr[0]
4     for num in arr[1:]:
5         if num > result:
6             result = num
7     return result
8
9 print(findMax([1, 2, 3, 4, 5]))
10 print(findMax([7, 7, 7, 7, 7]))
11 print(findMax([-10, 2, 3, -4, 5]))
```

5
7
5

=== Code Execution Successful ===

6. You have an algorithm that process a list of numbers. It firsts sorts the list using an efficient sorting algorithm and then finds the maximum element in sorted list. Write the code for the same.

Test Cases

1. Empty List

1. Input: []

2. Expected Output: None or an appropriate message indicating that the list is empty.
2. Single Element List
 1. Input: [5]
 2. Expected Output: 5
3. All Elements are the Same
 1. Input: [3, 3, 3, 3, 3]
 2. Expected Output: 3

The screenshot shows the Programiz Python Online Compiler interface. On the left, the code editor contains the following Python code:

```

1 def sort_and_find_max(nums):
2     if not nums:
3         return "List is empty"
4     nums.sort() # Timsort - O(n log n)
5     return nums[-1]
6
7 print(sort_and_find_max([]))
8 print(sort_and_find_max([5]))
9 print(sort_and_find_max([3, 3, 3, 3, 3]))
  
```

On the right, the output panel shows the results of the code execution:

```

List is empty
5
3
=== Code Execution Successful ===
  
```

7. Write a program that takes an input list of n numbers and creates a new list containing only the unique elements from the original list. What is the space complexity of the algorithm?

Test Cases

Some Duplicate Elements

- Input: [3, 7, 3, 5, 2, 5, 9, 2]
- Expected Output: [3, 7, 5, 2, 9] (Order may vary based on the algorithm used)

Negative and Positive Numbers

- Input: [-1, 2, -1, 3, 2, -2]
- Expected Output: [-1, 2, 3, -2] (Order may vary)

List with Large Numbers

- Input: [1000000, 999999, 1000000]
- Expected Output: [1000000, 999999]

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Output

Clear

```

1 def unique_elements(nums):
2     seen = set()
3     result = []
4     for num in nums:
5         if num not in seen:
6             seen.add(num)
7             result.append(num)
8     return result
9     # Space complexity: O(n)
10
11 print(unique_elements([3, 7, 3, 5, 2, 5, 9, 2]))
12 print(unique_elements([-1, 2, -1, 3, 2, -2]))
13 print(unique_elements([1000000, 999999, 1000000]))

```

```

[3, 7, 5, 2, 9]
[-1, 2, 3, -2]
[1000000, 999999]

=== Code Execution Successful ===

```

8. Sort an array of integers using the bubble sort technique. Analyze its time complexity using Big-O notation. Write the code

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Output

Clear

```

1 def bubble_sort(arr):
2     n = len(arr)
3     for i in range(n - 1):
4         for j in range(n - 1 - i):
5             if arr[j] > arr[j + 1]:
6                 arr[j], arr[j + 1] = arr[j + 1], arr[j]
7     return arr
8     # Time complexity: O(n^2)
9
10 arr = [5, 1, 4, 2, 8, 3]
11 print(bubble_sort(arr))

```

```

[1, 2, 3, 4, 5, 8]

=== Code Execution Successful ===

```

9. Checks if a given number x exists in a sorted array arr using binary search. Analyze its time complexity using Big-O notation.

Test Case:

Example X={ 3,4,6,-9,10,8,9,30} KEY=10

Output: Element 10 is found at position 5

Example X={ 3,4,6,-9,10,8,9,30} KEY=100

Output : Element 100 is not found

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```
1 def binary_search(arr, x):
2     # Time complexity: O(log n)
3     low, high = 0, len(arr) - 1
4     while low <= high:
5         mid = (low + high) // 2
6         if arr[mid] == x:
7             return mid
8         elif arr[mid] < x:
9             low = mid + 1
10        else:
11            high = mid - 1
12    return -1
13
14 arr = sorted([3, 4, 6, -9, 10, 8, 9, 30])
15 key = 10
16 pos = binary_search(arr, key)
17 print(f"Element {key} is found at position {pos}")
18
19 key = 100
20 pos = binary_search(arr, key)
21 print(f"Element {key} is not found")
```

Output

Clear

Element 10 is found at position 6
Element 100 is not found

=== Code Execution Successful ===

10. Given an array of integers nums, sort the array in ascending order and return it. You must solve the problem without using any built-in functions in $O(n \log(n))$ time complexity and with the smallest space complexity possible.

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```
1 def merge_sort(nums):
2     if len(nums) <= 1:
3         return nums
4     mid = len(nums) // 2
5     left = merge_sort(nums[:mid])
6     right = merge_sort(nums[mid:])
7     return merge(left, right)
8
9 def merge(left, right):
10    result = []
11    i = j = 0
12    while i < len(left) and j < len(right):
13        if left[i] <= right[j]:
14            result.append(left[i]); i += 1
15        else:
16            result.append(right[j]); j += 1
17    result.extend(left[i:])
18    result.extend(right[j:])
19    return result
20
21 nums = [5, 2, 3, 1, 4]
22 print(merge_sort(nums))
```

Output

Clear

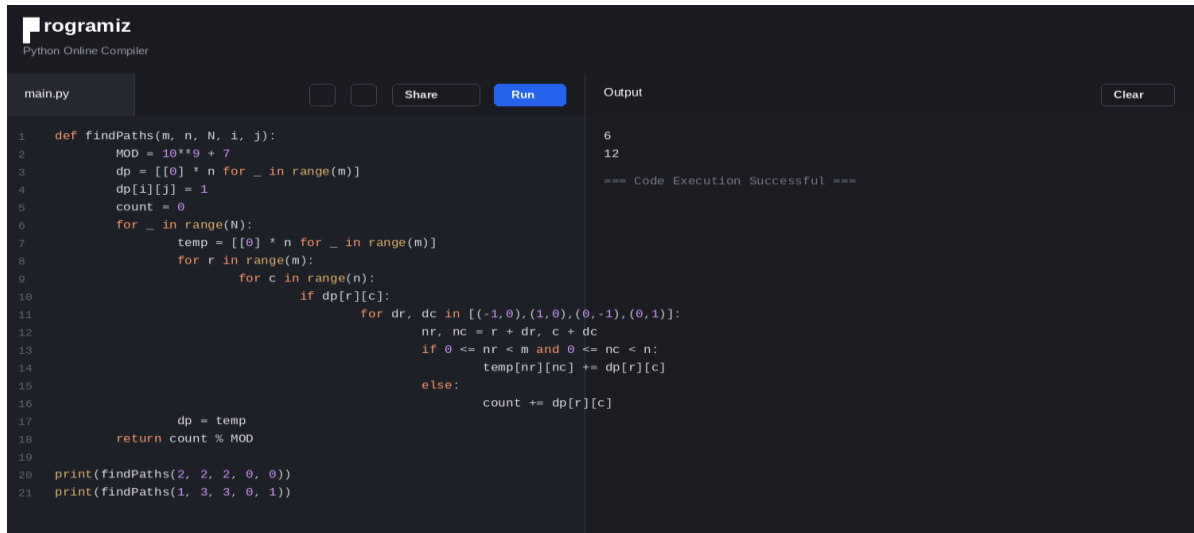
[1, 2, 3, 4, 5]

=== Code Execution Successful ===

11. Given an $m \times n$ grid and a ball at a starting cell, find the number of ways to move the ball out of the grid boundary in exactly N steps.

Example:

- Input: $m=2, n=2, N=2, i=0, j=0$ · Output: 6
- Input: $m=1, n=3, N=3, i=0, j=1$ · Output: 12



```
1 def findPaths(m, n, N, i, j):
2     MOD = 10**9 + 7
3     dp = [[0] * n for _ in range(m)]
4     dp[i][j] = 1
5     count = 0
6     for _ in range(N):
7         temp = [[0] * n for _ in range(m)]
8         for r in range(m):
9             for c in range(n):
10                if dp[r][c]:
11                    for dr, dc in [(-1,0), (1,0), (0,-1), (0,1)]:
12                        nr, nc = r + dr, c + dc
13                        if 0 <= nr < m and 0 <= nc < n:
14                            temp[nr][nc] += dp[r][c]
15                    else:
16                        count += dp[r][c]
17         dp = temp
18     return count % MOD
19
20 print(findPaths(2, 2, 2, 0, 0))
21 print(findPaths(1, 3, 3, 0, 1))
```

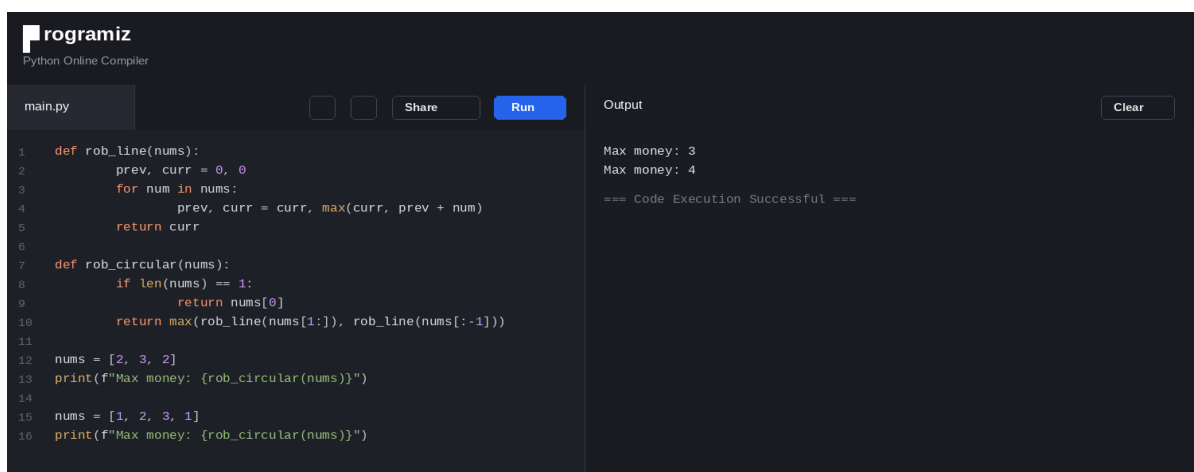
Output

```
6
12
=== Code Execution Successful ===
```

12. You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed. All houses at this place are arranged in a circle. That means the first house is the neighbor of the last one. Meanwhile, adjacent houses have security systems connected, and it will automatically contact the police if two adjacent houses were broken into on the same night.

Examples:

- (i) Input : $\text{nums} = [2, 3, 2]$
Output : The maximum money you can rob without alerting the police is 3 (robbing house 1).
- (ii) Input : $\text{nums} = [1, 2, 3, 1]$
Output : The maximum money you can rob without alerting the police is 4 (robbing house 1 and house 3).



```
1 def rob_line(nums):
2     prev, curr = 0, 0
3     for num in nums:
4         prev, curr = curr, max(curr, prev + num)
5     return curr
6
7 def rob_circular(nums):
8     if len(nums) == 1:
9         return nums[0]
10    return max(rob_line(nums[1:]), rob_line(nums[:-1]))
11
12 nums = [2, 3, 2]
13 print(f"Max money: {rob_circular(nums)}")
14
15 nums = [1, 2, 3, 1]
16 print(f"Max money: {rob_circular(nums)}")
```

Output

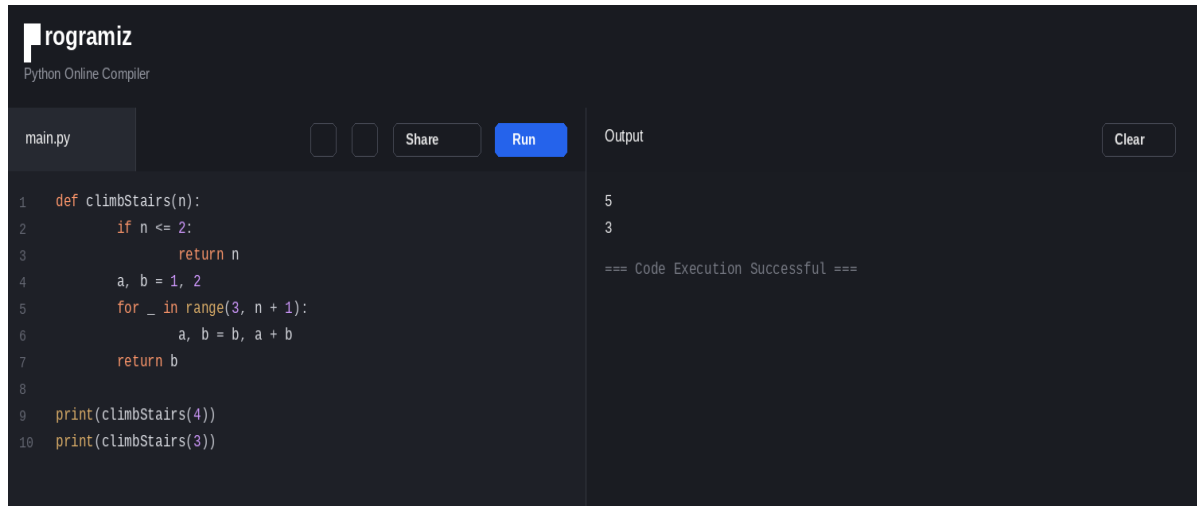
```
Max money: 3
Max money: 4
=== Code Execution Successful ===
```


13. You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Examples:

(i) Input: $n=4$ Output: 5

(ii) Input: $n=3$ Output: 3



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains the following Python code:

```
1 def climbStairs(n):
2     if n <= 2:
3         return n
4     a, b = 1, 2
5     for _ in range(3, n + 1):
6         a, b = b, a + b
7     return b
8
9 print(climbStairs(4))
10 print(climbStairs(3))
```

The output panel on the right shows the results of the code execution:

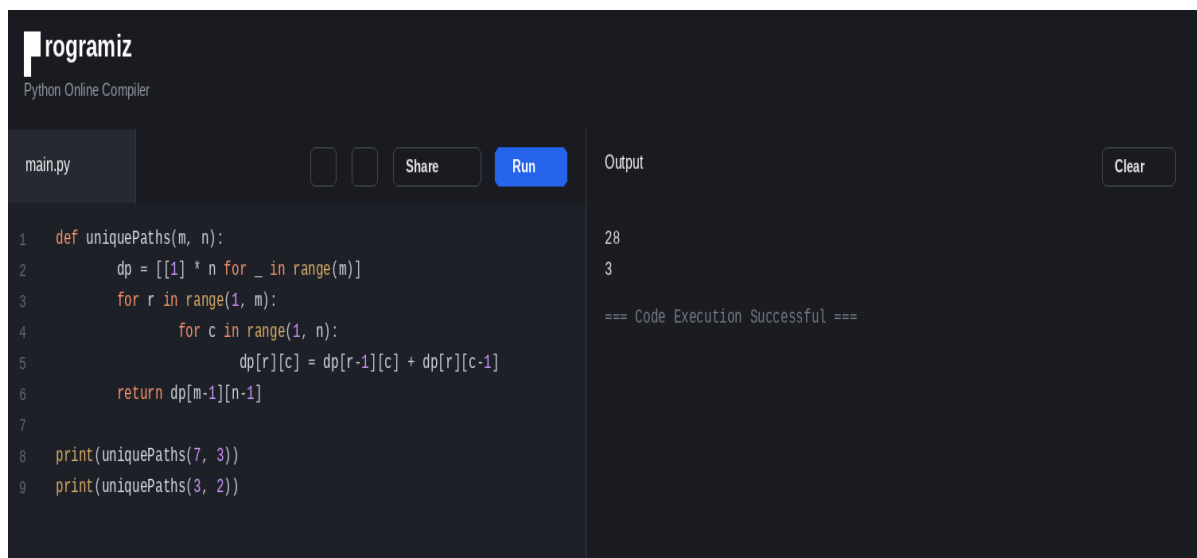
```
5
3
=== Code Execution Successful ===
```

14. A robot is located at the top-left corner of a $m \times n$ grid. The robot can only move either down or right at any point in time. The robot is trying to reach the bottom-right corner of the grid. How many possible unique paths are there?

Examples:

(i) Input: $m=7, n=3$ Output: 28

(ii) Input: $m=3, n=2$ Output: 3



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains the following Python code:

```
1 def uniquePaths(m, n):
2     dp = [[1] * n for _ in range(m)]
3     for r in range(1, m):
4         for c in range(1, n):
5             dp[r][c] = dp[r-1][c] + dp[r][c-1]
6     return dp[m-1][n-1]
7
8 print(uniquePaths(7, 3))
9 print(uniquePaths(3, 2))
```

The output panel on the right shows the results of the code execution:

```
28
3
=== Code Execution Successful ===
```

15. In a string *S* of lowercase letters, these letters form consecutive groups of the same character. For example, a string like *s* = "abbxxxxzzy" has the groups "a", "bb", "xxxx", "z", and "yy". A group is identified by an interval [start, end], where start and end denote the start and end indices (inclusive) of the group. In the above example, "xxxx" has the interval [3,6]. A group is considered large if it has 3 or more characters. Return the intervals of every large group sorted in increasing order by start index.

Example 1:

Input: *s* = "abbxxxxzzy"

Output: [[3,6]]

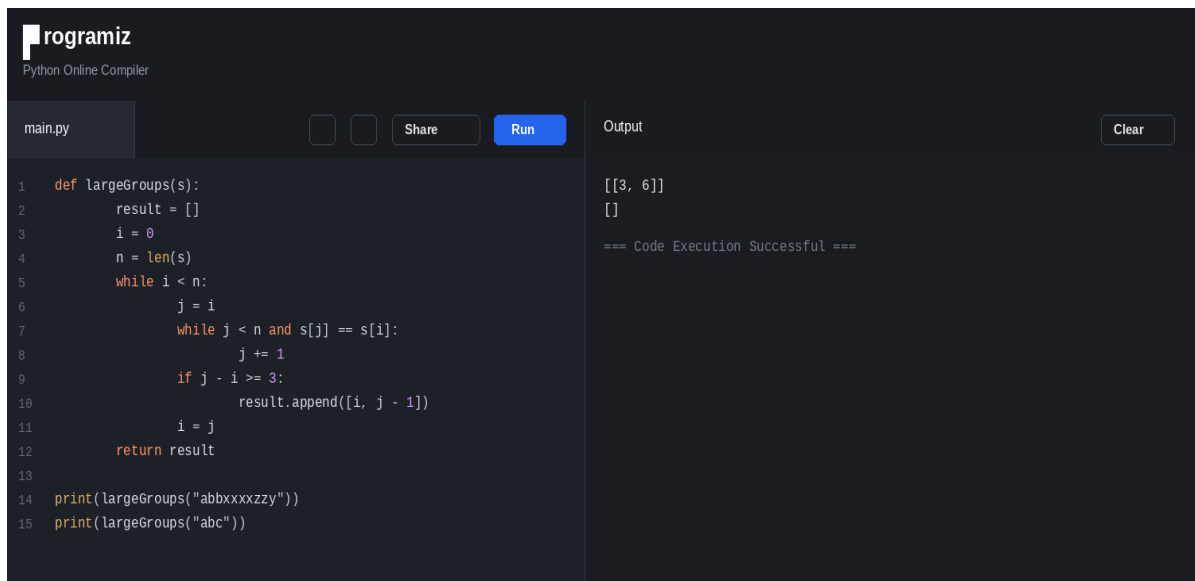
Explanation: "xxxx" is the only large group with start index 3 and end index 6.

Example 2:

Input: *s* = "abc"

Output: []

Explanation: We have groups "a", "b", and "c", none of which are large groups.



The screenshot shows a web-based Python compiler interface. On the left, there's a code editor with a file named 'main.py'. The code defines a function 'largeGroups(s)' that iterates through the string 's' to find groups of identical characters. It uses a while loop to extend the current group until a different character is found. If the group length is 3 or more, it appends the start and end indices to a result list. The code also includes print statements to verify the function's output for the examples 'abbxxxxzzy' and 'abc'. On the right, the 'Output' section shows the results: '[[3, 6]]' for the first example and '[]' for the second, followed by a success message '=== Code Execution Successful ==='.

```
1 def largeGroups(s):
2     result = []
3     i = 0
4     n = len(s)
5     while i < n:
6         j = i
7         while j < n and s[j] == s[i]:
8             j += 1
9         if j - i >= 3:
10            result.append([i, j - 1])
11        i = j
12    return result
13
14 print(largeGroups("abbxxxxzzy"))
15 print(largeGroups("abc"))
```

Output

```
[[3, 6]]
[]
=== Code Execution Successful ===
```

16. "The Game of Life, also known simply as Life, is a cellular automaton devised by the British mathematician John Horton Conway in 1970." The board is made up of an *m* x *n* grid of cells, where each cell has an initial state: live (represented by a 1) or dead (represented by a 0). Each cell interacts with its eight neighbors (horizontal, vertical, diagonal) using the following four rules

Any live cell with fewer than two live neighbors dies as if caused by under-population.

1. Any live cell with two or three live neighbors lives on to the next generation.
2. Any live cell with more than three live neighbors dies, as if by over-population.
3. Any dead cell with exactly three live neighbors becomes a live cell, as if by reproduction.

The next state is created by applying the above rules simultaneously to every cell in the current state, where births and deaths occur simultaneously. Given the current state of the $m \times n$ grid board, return *the next state*.

Example 1:

0	1	0
0	0	1
1	1	1
0	0	0

→

0	0	0
1	0	1
0	1	1
0	1	0

Input: board = [[0,1,0],[0,0,1],[1,1,1],[0,0,0]]

Output: [[0,0,0],[1,0,1],[0,1,1],[0,1,0]]

Example 2:

1	1
1	0

→

1	1
1	1

Input: board = [[1,1],[1,0]]

Output: [[1,1],[1,1]]


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```

1  def gameOfLife(board):
2      m, n = len(board), len(board[0])
3      next_state = [[0] * n for _ in range(m)]
4      for r in range(m):
5          for c in range(n):
6              live = 0
7              for dr in (-1, 0, 1):
8                  for dc in (-1, 0, 1):
9                      if dr == 0 and dc == 0:
10                         continue
11                     nr, nc = r + dr, c + dc
12                     if 0 <= nr < m and 0 <= nc < n:
13                         live += board[nr][nc]
14             if board[r][c] == 1:
15                 next_state[r][c] = 1 if live in (2, 3) else 0
16             else:
17                 next_state[r][c] = 1 if live == 3 else 0
18         return next_state
19
20 board = [[0,1,0],[0,0,1],[1,1,1],[0,0,0]]
21 print(gameOfLife(board))

```

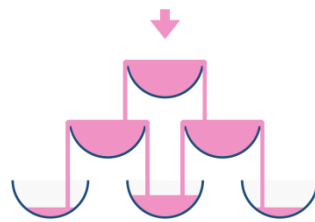
Output

[[0, 0, 0], [1, 0, 1], [0, 1, 1], [0, 1, 0]]

=== Code Execution Successful ===

Clear

17. We stack glasses in a pyramid, where the first row has 1 glass, the second row has 2 glasses, and so on until the 100th row. Each glass holds one cup of champagne. Then, some champagne is poured into the first glass at the top. When the topmost glass is full, any excess liquid poured will fall equally to the glass immediately to the left and right of it. When those glasses become full, any excess champagne will fall equally to the left and right of those glasses, and so on. (A glass at the bottom row has its excess champagne fall on the floor.) For example, after one cup of champagne is poured, the top most glass is full. After two cups of champagne are poured, the two glasses on the second row are half full. After three cups of champagne are poured, those two cups become full - there are 3 full glasses total now. After four cups of champagne are poured, the third row has the middle glass half full, and the two outside glasses are a quarter full, as pictured below.



Now after pouring some non-negative integer cups of champagne, return how full the j^{th} glass in the i^{th} row is (both i and j are 0-indexed.)

Example 1:

Input: poured = 1, query_row = 1, query_glass = 1

Output: 0.00000

Explanation: We poured 1 cup of champagne to the top glass of the tower (which is indexed as (0, 0)). There will be no excess liquid so all the glasses under the top glass will remain empty.

Example 2:

Input: poured = 2, query_row = 1, query_glass = 1

Output: 0.50000

Explanation: We poured 2 cups of champagne to the top glass of the tower (which is indexed as (0, 0)). There is one cup of excess liquid. The glass indexed as (1, 0) and the glass indexed as (1, 1) will share the excess liquid equally, and each will get half cup of champagne.

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```

1  def champagneTower(poured, query_row, query_glass):
2      tower = [[0.0] * k for k in range(1, 102)]
3      tower[0][0] = poured
4      for r in range(query_row + 1):
5          for c in range(len(tower[r])):
6              excess = (tower[r][c] - 1.0) / 2.0
7              if excess > 0:
8                  tower[r + 1][c] += excess
9                  tower[r + 1][c + 1] += excess
10         return min(1.0, tower[query_row][query_glass])
11
12     print(f"champagneTower(1, 1, 1):.5f")
13     print(f"champagneTower(2, 1, 1):.5f")

```

Output

0.00000

0.50000

=== Code Execution Successful ===

Clear